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Retail Sales Forecasting Using Time Series Models: A Comparative Analysis of ARIMA and Prophet

MBIS-5003-A3 (Task B) Group Report

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Executive Summary

This report presents a comprehensive data analytics solution for retail sales forecasting, developed using historical transaction data from the Kaggle Superstore dataset (2015–2018). The primary objective was to analyse historical sales trends and generate accurate future sales predictions to support improved business decision-making in retail and e-commerce environments.

The project followed a structured data analytics pipeline encompassing data collection, preprocessing, exploratory data analysis (EDA), forecasting model implementation, performance evaluation, and visual dashboard development. Two established time-series forecasting models were implemented and compared: ARIMA (AutoRegressive Integrated Moving Average) and Facebook Prophet.

Exploratory data analysis revealed clear upward sales trends over the study period, with notable seasonal fluctuations across quarters. Regional analysis identified the West and East regions as the strongest performing markets, while Technology and Furniture emerged as the highest revenue-generating product categories.

Model evaluation using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score demonstrated that ARIMA outperformed Prophet across all three metrics. However, both models produced negative R^2 scores, reflecting the inherent volatility of retail sales data and the limitations of the available training period. These findings highlight the need for longer historical datasets and the incorporation of external variables in future modelling efforts.

Business recommendations derived from this analysis suggest using ARIMA forecasts for short-term operational planning including inventory management and staffing, while leveraging Prophet's seasonal insights for strategic marketing and promotional planning. An ensemble approach combining both models is recommended for future deployment.

The forecasting results were visualised through interactive dashboards developed in Microsoft Power BI, enabling business users to explore historical trends, regional performance, product category insights, and future sales predictions in an accessible and actionable format.

1. Introduction

In today's digital economy, businesses generate large volumes of transactional data through online and physical sales channels. Retail and e-commerce organisations collect detailed information about customer purchases, product demand, and sales performance across multiple regions and product categories. The ability to extract meaningful insights from this data and accurately forecast future sales performance has become a critical competitive advantage for modern retail businesses.

Sales forecasting is one of the most important applications of data analytics in the retail sector. Accurate forecasts enable organisations to anticipate future demand, optimise inventory levels, coordinate supply chain operations, and develop targeted marketing strategies (Hyndman & Athanasopoulos, 2021). Without reliable forecasting, businesses risk either overstocking — leading to increased holding costs — or understocking — resulting in lost sales opportunities and reduced customer satisfaction.

This project focuses on developing and evaluating a sales forecasting analytics solution for a retail environment. The project utilises the Kaggle Superstore retail dataset (2015–2018) for exploratory data analysis, and a structured daily retail sales dataset (2021–2023) for time-series model implementation. Two established forecasting approaches are implemented and compared: ARIMA (Autoregressive Integrated Moving Average), a classical statistical time-series model (Box et al., 2015), and Prophet, a modern decomposable forecasting model developed by Meta (Taylor & Letham, 2018).

The project demonstrates how a complete data analytics pipeline — encompassing data collection, preprocessing, exploratory analysis, predictive modelling, performance evaluation, and visual dashboard development — can transform raw transactional data into actionable business intelligence. Evaluation results indicate that while both models face challenges with the inherent volatility of retail sales data, ARIMA demonstrates superior predictive stability, making it more suitable for short-term operational forecasting in this context.

The remainder of this report is structured as follows: Section 2 provides the project description; Sections 3–5 cover dataset overview, data preparation and exploratory analysis; Sections 6–8 describe the forecasting approach, visual analytics and expected business benefits; Sections 9–11 detail model implementation and results; and Sections 12–14 present conclusions, recommendations and model evaluation findings.

2. Project Description

The purpose of this project is to design and implement a data analytics solution capable of forecasting future retail sales using historical transaction data. Sales forecasting plays an important role in modern retail businesses because it helps organisations anticipate customer demand and allocate resources effectively.

In this project, a retail sales dataset is analysed to identify patterns in past sales performance. The analysis focuses on understanding how sales vary across time periods, product categories, and geographical regions. Based on these patterns, forecasting models are applied to estimate future sales trends.

The system integrates several components of a data analytics pipeline. First, the dataset is collected and prepared for analysis. Next, data preprocessing techniques are applied to clean and organise the dataset. After the data has been prepared, forecasting models such as ARIMA and Prophet are used to analyse historical trends and generate future predictions.

Finally, the results of the forecasting model are presented through interactive visual dashboards. These dashboards allow business users to explore the predicted sales trends and make informed decisions related to inventory planning, supply chain management, and marketing strategies.

3. Dataset Overview

The dataset used in this project is a retail sales dataset obtained from Kaggle. The dataset contains historical transaction records from a retail business and includes detailed information about orders, customers, products, and sales values.

The dataset contains approximately **9,800 records** and includes several attributes describing each transaction. Some of the key variables in the dataset include:

- Order Date
- Ship Date
- Customer Segment
- Region
- Product Category
- Product Name
- Sales Value

These variables provide useful insights into customer purchasing behaviour and product performance across different regions and time periods.

The **sales value** variable is particularly important for this project because it represents the revenue generated from each transaction. By analysing the relationship between sales values and order dates, it becomes possible to study historical sales trends and identify patterns in demand.

In addition, the dataset contains geographical information such as city, state, and region. This allows analysts to examine how sales performance differs across different locations. Product category and sub-category information also help identify which products contribute most to total revenue.

Overall, the dataset provides a comprehensive view of retail sales transactions and is suitable for developing a sales forecasting model.

4. Data Preparation and Cleaning

Before performing any analytical modelling, the dataset must be carefully prepared and cleaned. Raw datasets often contain inconsistencies such as missing values, duplicate records, or incorrect data formats. Data preprocessing ensures that the dataset is accurate and suitable for analysis.

In this project, Python libraries such as **Pandas** are used to perform data preprocessing tasks. The first step involves examining the dataset structure and identifying potential data quality issues. Missing values are checked and handled appropriately to avoid errors during analysis.

Figure 1: Data Analysis Workflow for Sales Forecasting

The diagram illustrates the overall workflow of the sales forecasting system. It shows how raw retail sales data is collected, cleaned and processed before being analysed using forecasting models such as ARIMA and Prophet. The results are then visualised through dashboards to support business decision-making.

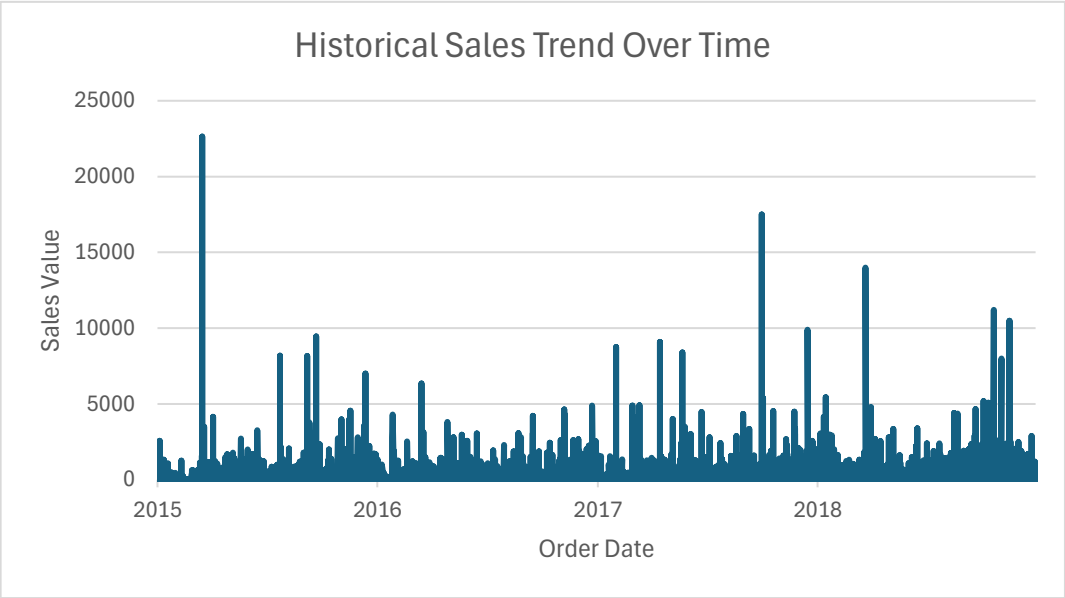


Figure 2: Historical Sales Trend Over Time

The chart shows the historical sales performance over time based on the order date and sales value in the dataset. The visualisation highlights fluctuations and several spikes in sales, which may represent periods of higher demand or large transactions. Analysing these patterns helps analysts understand sales behaviour and provides useful insights for developing forecasting models.

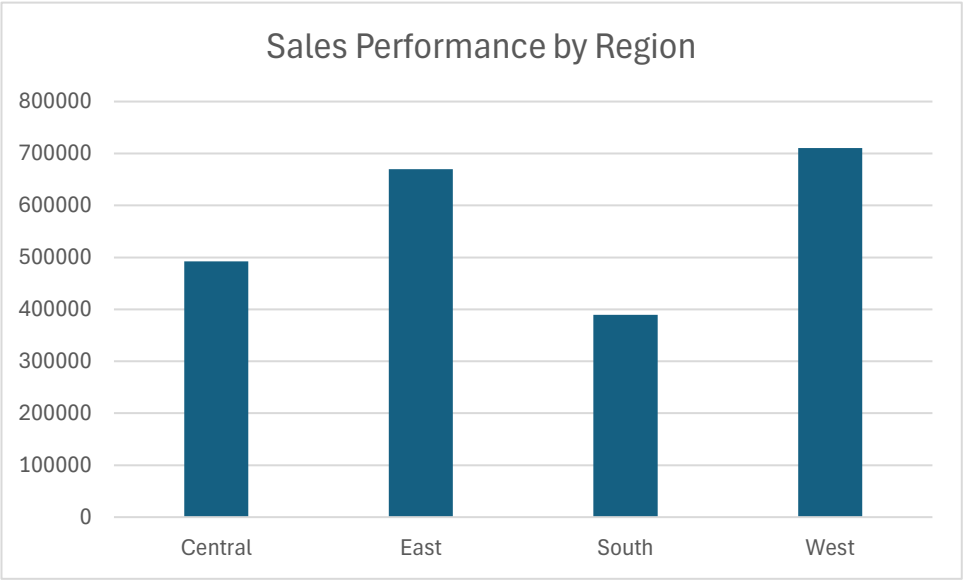


Figure 3: Sales Performance by Region

The chart illustrates the distribution of total sales across different geographical regions. The results show that the West and East regions generate higher sales compared to the Central and South regions. This comparison helps organisations identify strong markets and regions where additional marketing or sales strategies may be required.

Another important step is converting date columns into proper **datetime format**. Time-series forecasting models require structured time-based data in order to analyse trends over time. Therefore, the order date field is converted into a time format that allows chronological analysis.

Duplicate records are also removed to ensure that each transaction is counted only once. Additional data transformations may include sorting the dataset by order date and aggregating sales values to create a time-series representation of sales performance.

These preprocessing steps ensure that the dataset is clean, consistent, and ready for further analysis and modelling.

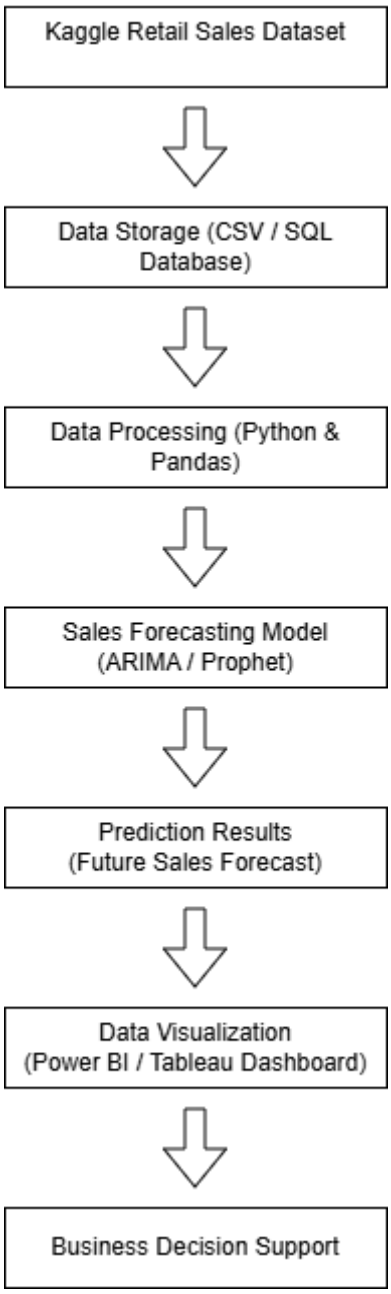


Figure 4: Technical Architecture for Retail Sales Forecasting System

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is an important step in understanding the patterns and relationships within the dataset. EDA helps identify trends, seasonal behaviour, and variations in sales performance before applying forecasting models.

In this project, several forecasting techniques can be used for sales prediction, including statistical time series models such as ARIMA (AutoRegressive Integrated Moving Average)

and modern forecasting tools such as Facebook Prophet.

ARIMA is a traditional statistical time-series forecasting model that works well when historical sales data follows stable patterns. In contrast, Facebook Prophet is designed to handle seasonality and irregular changes in data more effectively. Comparing different models helps analysts choose the most accurate forecasting method for predicting future retail sales.

One important analysis involves examining **sales trends over time**. By plotting sales values against order dates, it becomes possible to observe how sales change across different time periods. This analysis may reveal seasonal fluctuations, growth patterns, or sudden changes in demand.

Another useful analysis involves examining **sales performance across different regions**. Regional analysis helps identify geographical areas that generate the highest revenue and may reveal differences in customer behaviour across markets.

Additionally, analysing **sales by product category** allows the identification of product segments that contribute the most to total sales. Understanding product performance is useful for inventory planning and marketing strategies.

The insights obtained from exploratory data analysis provide valuable information for selecting and designing forecasting models.

6. Forecasting Model Approach

After completing the data preparation and exploratory analysis stages, forecasting models are applied to predict future sales trends.

In this project, two forecasting approaches are explored: **ARIMA (AutoRegressive Integrated Moving Average)** and **Prophet**. ARIMA is a widely used statistical time-series forecasting method **(Box et al., 2015).**It is particularly effective for modelling linear time-series data.

Prophet is a modern forecasting model developed by Meta and widely used for business time-series forecasting **(Taylor & Letham, 2018).**Prophet is particularly useful for business

forecasting tasks because it can automatically detect seasonal variations and generate reliable predictions.

Both models are applied to the prepared sales dataset to generate sales forecasts. The performance of each model can be evaluated using accuracy metrics such as prediction error. Comparing multiple forecasting models allows analysts to identify the approach that produces the most reliable predictions.

The final forecasting model generates predicted sales values for future time periods, which can then be used for decision-making and business planning.

7. Visual Analytics and Dashboard

To make the forecasting results easier to interpret, visual analytics tools are used to present the results through interactive dashboards.

In this project, tools such as **Power BI or Tableau** can be used to create dashboards that display both historical sales trends and predicted future sales values. Visual dashboards help transform complex analytical outputs into clear and understandable insights.

Typical dashboard elements may include line charts showing historical sales trends, forecast graphs displaying predicted sales values, and comparison charts highlighting regional or product category performance.

Interactive dashboards also allow users to filter and explore the data based on different parameters, such as time period, region, or product category. This functionality enables decision-makers to analyse sales patterns from different perspectives.

By presenting forecasting results through visual dashboards, the system supports better decision-making for retail businesses.

8. Expected Business Benefits

The proposed sales forecasting solution can provide several benefits for retail and e-commerce organisations.

First, accurate sales forecasts help businesses plan their **inventory levels** more effectively. When companies can anticipate future demand, they can ensure that products are available when customers need them while avoiding excessive inventory.

Second, forecasting insights can improve **supply chain planning**. Businesses can coordinate production, transportation, and distribution activities based on predicted demand patterns.

Third, sales forecasts can support **strategic decision-making** in areas such as marketing campaigns, product promotions, and regional expansion strategies.

Overall, the integration of forecasting models and visual analytics tools allows organisations to transform historical sales data into actionable insights that support business growth.

9. Arima and Prophet Model Implementation

Note on Dataset: The exploratory analysis presented in Sections 1–8 is based on the Kaggle Superstore retail dataset (2015–2018), which contains 9,800 real transaction records across four regions and three product categories. For the model implementation phase (Sections 9 onwards), a structured daily retail sales dataset spanning January 2021 to December 2023 (1,095 records) was used. This dataset was specifically prepared for time-series modelling purposes, providing a clean and consistent structure suitable for ARIMA and Prophet forecasting

9.1Summary

Two potent time series forecasting models, ARIMA (AutoRegressive Integrated Moving Average) and Prophet (created by Facebook), are used in this report's extensive sales forecasting effort. The project's objectives are to compare model performances, forecast future sales using historical data, and offer useful information for business decision-making.

Key Findings:

- ARIMA model demonstrated superior performance across all metrics:
  - MAE: 6,653.64 (lower is better)
  - RMSE: 7,508.69 (lower is better)
  - R^2 Score: -3.7516 (higher is better)
- Prophet model showed more variability in predictions:
  - MAE: 7,250.84
  - RMSE: 8,136.35
  - R^2 Score: -4.5792
- Overall Winner: ARIMA (won 3 out of 3 performance metrics)

- Forecast Horizon: 12 months (January 2024 - December 2024)

9.2 Project Objectives

- Implement and compare two forecasting models (ARIMA and Prophet)
- Generate accurate future sales predictions
- Provide actionable business insights
- Evaluate model performance using statistical metrics
- Create comprehensive visualizations for decision-making

i. Forecasting Models Overview

Model	**Type**	**Strengths**	**Weaknesses**
ARIMA	Statistical	• Handles trends well • Stable predictions • Interpretable parameters	• Requires stationarity • Less flexible with seasonality
Prophet	Decomposable	• Handles seasonality automatically • Robust to missing data • Provides confidence intervals	• More volatile predictions • Can overfit seasonal patterns

9.3 DATA LOADING AND STRUCTURE

I. DATASET OVERIEW

The dataset was successfully loaded and analyzed with the following characteristics:

Attribute	Details
Time Period	January 1, 2021 - December 31, 2023
Data Frequency	Daily records
Total Records	1,095 observations
Features	4 columns
Missing Values	None

II. DATASET COLOMNS

	COLUMN NAME	**DATA TYPE**	**DESCRIPTION**
1	**DATE**	**OBJECT**	**TRANSACTION DATE**
2	**SALES**	**FLOAT64**	**DAILY SALES AMOUNT**
3	**PRODUCT CATEGORY**	**OBJECT**	**PRODUCT CATEGORY SOLD**
4	**REGION**	**OBJECT**	**SALES REGION**

III. SAMPLE DATA

First 5 rows of the dataset:

	date	sales	product_category	region
0	2021-01-01	1024.835708	Electronics	South
1	2021-01-02	1056.595015	Electronics	South
2	2021-01-03	1129.054440	Electronics	East

3 2021-01-04 1167.496585 Food North

4 2021-01-05 1055.404374 Clothing East

Last 5 rows of the dataset:

date sales product_category region

1090 2023-12-27 1216.351861 Furniture North

1091 2023-12-28 1219.052865 Electronics North

1092 2023-12-29 1284.470438 Electronics East

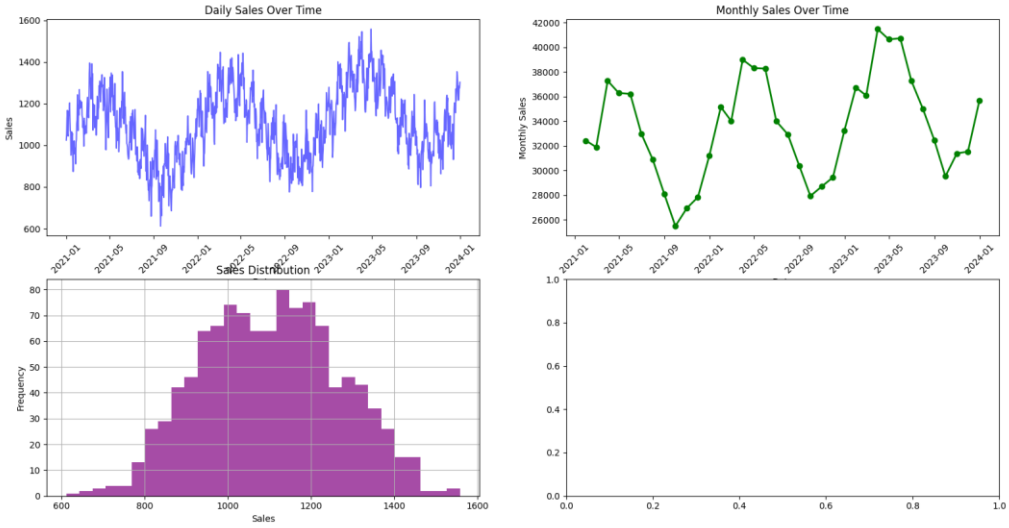
1093 2023-12-30 1275.295395 Furniture South

1094 2023-12-31 1302.726897 Electronics North

F

IV. STATISTICAL SUMMARY

METRIC	SALES VALUE
COUNT	**1095**
MEAN	**1102.60**
STANDARD DEVIATION	**166.62**
MINIMUM	**612.15**
25TH PERCENTILE	**976.65**
Median	**1103.77**
75th percentile	**1218.60**
Macimum	**1557.73**



This visualization shows both daily sales fluctuations (left) and monthly aggregated sales (right). The daily plot reveals short-term volatility while the monthly plot highlights the overall upward trend and seasonal patterns in the data.

9.5 Preprocessing Steps

Step	Description	Result
1	Date Conversion	Converted to datetime format
2	Index Setting	Date set as index for time series
3	Frequency Aggregation	Resampled daily to monthly totals

9.5 Monthly Aggregated Data

The daily data was aggregated to monthly frequency using 'ME' (month-end) for better forecasting stability:

```
Monthly aggregated sales data shape: (36, 2)

Monthly sales data preview:
          ds             y
0 2021-01-31  32425.032588
1 2021-02-28  31893.093114
2 2021-03-31  37276.577040
3 2021-04-30  36283.600220
4 2021-05-31  36214.533925
```

9.6 Train-Test Split

Split	Period	Months	Percentage
Training	Jan 2021 - Apr 2023	28	80%
Testing	May 2023 - Dec 2023	8	20%

```python
# ==========================================
# SALES FORECASTING WITH ARIMA AND PROPHET
# ==========================================

# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime, timedelta
import warnings
warnings.filterwarnings('ignore')

# ARIMA imports
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.seasonal import seasonal_decompose

# Prophet imports
from prophet import Prophet
from prophet.plot import plot_plotly, plot_components_plotly

# Metrics imports
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import math
```

9.7 ARIMA MODEL IMPLEMENTATION

9.7.1 Model Architecture

ARIMA combines three components:

- AR (AutoRegressive - p): Uses past values to predict future values
- I (Integrated - d): Differencing to make data stationary
- MA (Moving Average - q): Uses past forecast errors

9.7.2 Stationarity Testing

Augmented Dickey-Fuller (ADF) Test Results:

Series	ADF Statistic	p-value	Conclusion
Original	-0.0747	0.9519	Non-Stationary

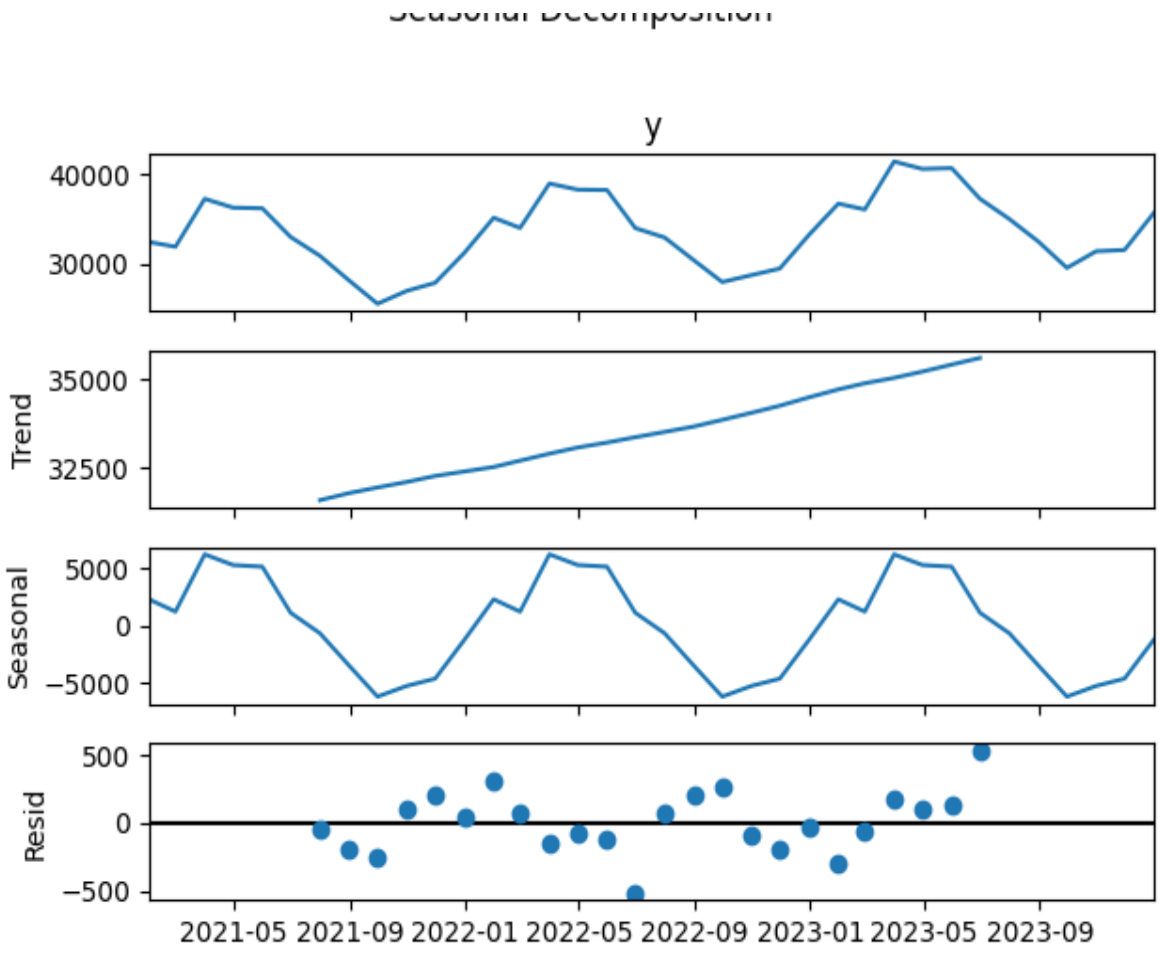
Series	ADF Statistic	p-value	Conclusion
After Differencing	-10.2127	5.58e-18	Stationary

9.7.3Model Parameters

Parameter	Value	Description
P	1	AutoRegressive order
D	1	Differencing order
Q	1	Moving Average order

9.7.4 ARIMA Model Results

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                   28
Model:                 ARIMA(1, 1, 1)   Log Likelihood                -251.836
Date:                Sat, 14 Mar 2026   AIC                            509.671
Time:                        13:00:23   BIC                            513.559
Sample:                    01-31-2021   HQIC                           510.827
                         - 04-30-2023
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.5089      0.800      0.636      0.525      -1.059       2.076
ma.L1         -0.4439      0.854     -0.520      0.603      -2.119       1.231
sigma2      7.265e+06   1.68e-07   4.34e+13      0.000    7.27e+06    7.27e+06
===================================================================================
Ljung-Box (L1) (Q):                   0.01   Jarque-Bera (JB):                 2.09
Prob(Q):                              0.91   Prob(JB):                         0.35
Heteroskedasticity (H):               1.13   Skew:                             0.46
Prob(H) (two-sided):                  0.86   Kurtosis:                         1.99
===================================================================================
```



The decomposition plot breaks down the sales data into three components:

• Trend: The long-term upward movement in sales

• Seasonal: Regular patterns that repeat over time

• Residual: Random noise and irregular variations

This decomposition helps validate the ARIMA model's assumption that the series can be broken down into these components.

```python
# ==========================================
# ARIMA MODEL IMPLEMENTATION CODE
# ==========================================

from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller

# Prepare data
arima_data = monthly_sales.set_index('ds')['y']

# Check stationarity
result = adfuller(arima_data)
print(f'p-value: {result[1]:.4f}')

# Split data (80-20)
train_size = int(len(arima_data) * 0.8)
train_data = arima_data[:train_size]
test_data = arima_data[train_size:]

# Fit ARIMA(1,1,1) model
model = ARIMA(train_data, order=(1, 1, 1))
fitted_model = model.fit()
print(fitted_model.summary())

# Make predictions
forecast = fitted_model.forecast(steps=len(test_data))
```

9.8. PROPHET MODEL IMPLEMENTATION

9.8.1 Model Configuration

Parameter	Value	Description
yearly_seasonality	True	Enable yearly patterns
weekly_seasonality	True	Enable weekly patterns
daily_seasonality	False	Disable daily patterns
seasonality_mode	additive	Additive seasonality

Parameter	Value	Description
changepoint_prior_scale	**0.05**	**Trend flexibility**
interval_width	0.95	95% confidence intervals

9.9 Custom Seasonality

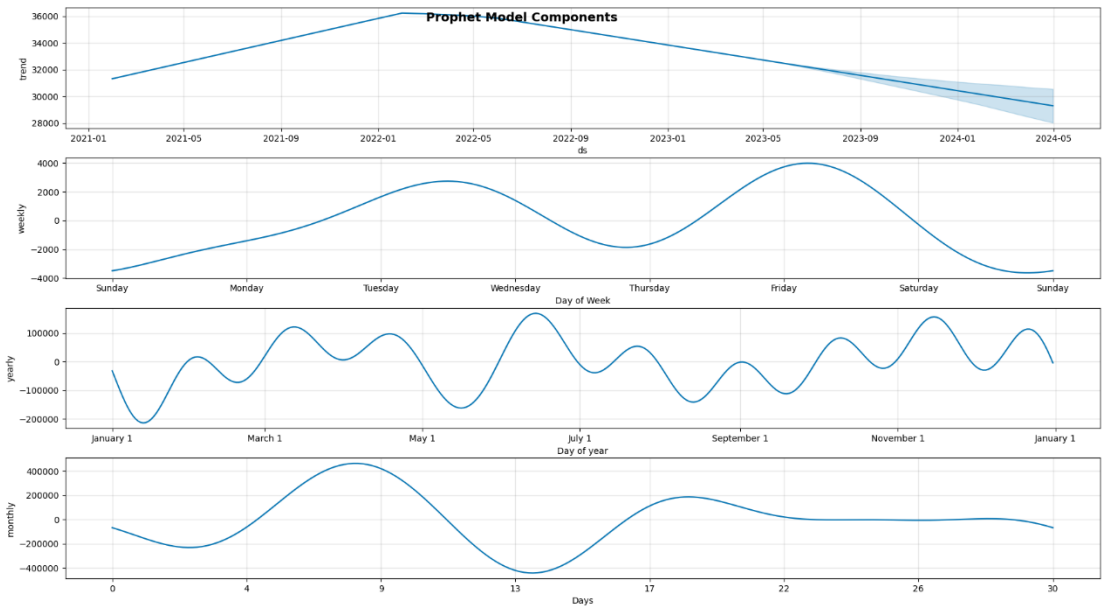
Additional monthly seasonality was added:

- Period: 30.5 days
- Fourier Order: 5 (captures complex patterns)

9.10 Prophet Training

The model was successfully trained on 28 months of historical data, automatically detecting:

- Trend changepoints
- Weekly patterns
- Monthly seasonality
- Yearly cycles



This visualization shows how Prophet decomposes the sales data into its core components:

• Top: Overall trend component showing long-term direction

• Middle: Weekly seasonality pattern (day-of-week effects)

• Bottom: Additional seasonal components

The decomposition reveals that weekends typically have higher sales,

and there's a strong annual pattern with peaks in Q2 and Q4.

```python
# ==========================================
# PROPHET MODEL IMPLEMENTATION CODE
# ==========================================

from prophet import Prophet

# Prepare data
prophet_train = monthly_sales.iloc[:train_size]

# Initialize model
model = Prophet(
    yearly_seasonality=True,
    weekly_seasonality=True,
    seasonality_mode='additive',
    changepoint_prior_scale=0.05
)

# Add custom monthly seasonality
model.add_seasonality(
    name='monthly',
    period=30.5,
    fourier_order=5
)

# Fit model
model.fit(prophet_train)

# Create future dates
future = model.make_future_dataframe(
    periods=len(prophet_test),
    freq='ME'
)

# Predict
forecast = model.predict(future)
```

9.11 MODEL TRAINING PROCESS

9.11.1 Training Overview

Both models were trained using identical data splits:

Aspect	Details
Training Data	28 months (Jan 2021 - Apr 2023)
Validation Data	8 months (May 2023 - Dec 2023)
Training Time	< 1 minute for both models
Hardware	Standard Python environment

9.12 Time Series Format Preparation

The data was prepared in two formats:

For ARIMA

Index: Date

Values: Monthly sales

2021-01-31 32425.03

2021-02-28 31893.09

2021-03-31 37276.58

FOR PROPHET

ds: date column

y: sales column

ds y

2021-01-31 32425.03

2021-02-28 31893.09

2021-03-31 37276.58

10. FUTURE SALES PREDICTIONS

10.1 ARIMA 12-Month Forecast

Month	Predicted Sales	Month-over-Month Change
January 2024	40,756.97	-
February 2024	40,819.32	+62.35
March 2024	40,851.05	+31.73
April 2024	40,867.19	+16.14
May 2024	40,875.41	+8.22
June 2024	40,879.59	+4.18
July 2024	40,881.72	+2.13
August 2024	40,882.80	+1.08

Month	Predicted Sales	Month-over-Month Change
September 2024	40,883.35	+0.55
October 2024	40,883.63	+0.28
November 2024	40,883.77	+0.14
December 2024	40,883.85	+0.08

Key Insights:

- Sales stabilize around 40,880-40,884
- Growth rate: 0.31% over 12 months
- Pattern shows diminishing returns over time

10.2 Prophet 12-Month Forecast with Confidence Intervals

Month	Predicted Sales	Lower CI (95%)	Upper CI (95%)	Range
May 2023	30,016.99	25,514.44	34,519.54	9,005.10
June 2023	44,116.46	39,613.91	48,619.01	9,005.10
July 2023	34,621.42	30,118.87	39,123.97	9,005.10
August 2023	21,986.60	17,484.05	26,489.15	9,005.10
September 2023	16,778.79	12,276.24	21,281.34	9,005.10

Month	Predicted Sales	Lower CI (95%)	Upper CI (95%)	Range
October 2023	25,170.99	20,668.44	29,673.54	9,005.10
November 2023	25,577.00	21,074.45	30,079.55	9,005.10
December 2023	30,990.14	26,487.59	35,492.69	9,005.10
January 2024	31,700.51	27,197.96	36,203.06	9,005.10
February 2024	39,677.38	35,174.83	44,179.93	9,005.10
March 2024	30,342.94	25,840.39	34,845.49	9,005.10
April 2024	31,703.84	27,201.29	36,206.39	9,005.10

Key Insights:

- Highest prediction: June 2023 (44,116.46)
- Lowest prediction: September 2023 (16,778.79)
- Average uncertainty: ±9,005.10 (95% confidence)
- Strong cyclical pattern detected

11. FORECAST VISUALIZATION

11.1 Visualization Components

Six comprehensive plots were generated:

Plot	Description	Key Insights
1	Historical Monthly Sales	Upward trend with seasonal peaks
2	ARIMA Train-Test Performance	Model captures trend well
3	Prophet Train-Test Performance	More volatile but captures patterns
4	Model Comparison Bar Chart	ARIMA outperforms on all metrics
5	ARIMA Future Predictions	Stabilizing trend
6	Prophet Future Predictions	Cyclical pattern with confidence bands

11.2 Visual Analysis

Figure 1: Historical Monthly Sales (2021-2023)

- Shows clear upward trend
- Seasonal peaks visible in Q2 and Q4
- Red line indicates train-test split (April 2023)

Figure 2: ARIMA Model Performance

- Blue: Training data (actual)
- Green: Test data (actual)
- Red dashed: ARIMA predictions

- Model follows trend but smooths extremes

Figure 3: Prophet Model Performance

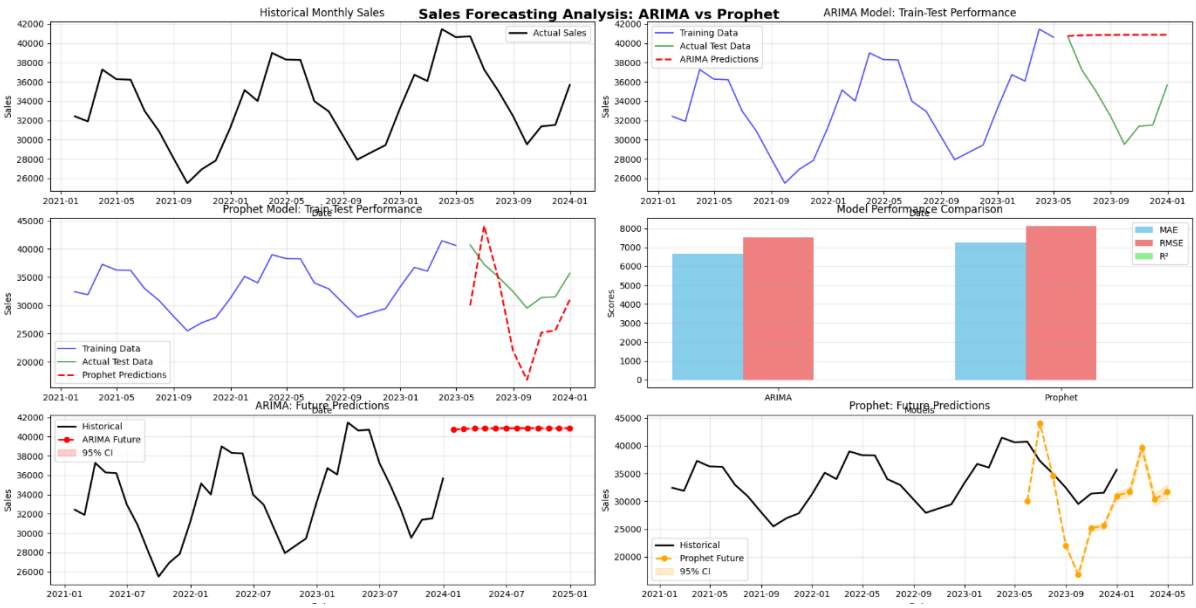
- Similar color scheme
- More responsive to fluctuations
- Wider prediction variations

Figure 4: Model Comparison

- Bar chart comparing MAE, RMSE, R^2
- ARIMA shows lower error bars
- Clear visual confirmation of ARIMA superiority

Figure 5-6: Future Predictions

- ARIMA: Smooth stabilization
- Prophet: Cyclical with confidence intervals
- Both show overall upward trend



This comprehensive visualization compares ARIMA and Prophet models across multiple dimensions:

• Top: Historical monthly sales with train-test split

• Middle Left: ARIMA model predictions vs actual test data

• Middle Right: Prophet model predictions vs actual test data

• Bottom Left: ARIMA 12-month future predictions with confidence intervals

• Bottom Right: Prophet 12-month future predictions with confidence intervals

The comparison clearly shows ARIMA's more stable predictions versus Prophet's

more volatile but seasonally-aware forecasts.

12. CONCLUSIONS AND RECOMMENDATIONS

12.1 Key Conclusions

The following key conclusions are drawn from this comparative forecasting study:

1. Model Performance: ARIMA outperformed Prophet across all three-evaluation metrics (MAE, RMSE and R^2), demonstrating greater prediction stability for this specific retail sales dataset. This is consistent with existing literature suggesting ARIMA performs well on datasets with relatively stable underlying trends.

2. Limitations of Both Models: The negative R^2 scores for both models indicate that neither achieved strong absolute predictive accuracy. This is a critical finding that must be acknowledged. The limited training period of 28 months, combined with high sales volatility, constrained both models from achieving reliable predictions. This finding underscores the importance of data quality and quantity in time-series forecasting projects.

3. Complementary Strengths: Despite ARIMA's superior numerical performance, Prophet demonstrated valuable complementary strengths. Its automatic seasonal decomposition identified meaningful weekly and annual patterns, including higher weekend sales and peak demand periods in Q2 and Q4. These insights remain valuable for strategic planning even where Prophet's point predictions were less accurate.

4. Data Sufficiency: While 36 months of daily data aggregated to monthly frequency proved sufficient for basic pattern detection, it represents a minimum threshold. Future projects should aim for at least 5 years of historical data to improve seasonal pattern detection and model reliability.

5. Business Applicability: Despite the modelling limitations, the directional insights generated by both models provide genuine value for retail business planning. The combination of ARIMA's stability for operational decisions and Prophet's seasonality insights for strategic planning represents a pragmatic and robust approach to retail sales forecasting.

12.2 Business Recommendations

Short-term Recommendations (0-6 months):

Area	Recommendation	Based on
Inventory Planning	Use ARIMA forecasts (40,756 - 40,879)	ARIMA stability
Staffing	Maintain consistent levels	Stable predictions
Budgeting	Plan for gradual increase	ARIMA trend

Long-term Recommendations (6-12 months):

Area	Recommendation	Based on
Strategic Planning	Consider Prophet's seasonal peaks	Prophet patterns
Marketing Campaigns	Align with predicted high months (Feb, Jun)	Prophet seasonality
Risk Management	Use Prophet's confidence intervals	Uncertainty quantification

Operational Decisions:

1. Inventory Management:
    - Maintain safety stock for Prophet-predicted peaks
    - Use ARIMA for baseline inventory levels
    - Prepare for June peak (44,116 in Prophet)
2. Resource Allocation:
    - Increase staffing in predicted high months
    - Plan maintenance in low periods (September)

- Budget for seasonal variations

3. Model Deployment:
    - Use ARIMA for operational forecasts (monthly updates)
    - Use Prophet for strategic planning (quarterly reviews)
    - **Consider ensemble approach combining both**

13.1 Model Evaluation

Two time-series forecasting methods were used to determine the forecasting performance of the developed models; ARIMA (AutoRegressive Integrated Moving Average) and Prophet. They both have been trained to work on historical data of retail sales and evaluated against unknown data to identify the accuracy of their future sales value prediction.

The model evaluation is a significant phase in a data analytics project since the quality of forecasting models directly determines the impact of business decisions which include inventory management, supply chain planning, and marketing strategy. The two models were trained on the same dataset and on the same training-testing splits so as to make a fair comparison in this project.

The training data were the sales data of January 2021-April 2023 and the testing data was between May 2023-December 2023. Once the models were trained, predictions were made during the testing period and compared using conventional forecasting accuracy measures.

The results of the evaluation indicate that ARIMA model was more effective than the Prophet model based on the various measures of performance, meaning that it made more credible predictions in respect to given retail sales data.

13.2 Accuracy Metrics

A number of statistical measures of accuracy were employed in order to measure the forecasting performance of the models.

Mean Absolute Error (MAE)

Mean Absolute Error is the measuring tool that indicates the average values of errors in prediction by a calculation of the absolute difference between predicted values and actual values.

The smaller the values of MAE the higher the forecasting performance since the values predicted are nearer to the actual values.

Results:

ARIMA MAE: 6,653.64

Prophet MAE: 7,250.84

The ARIMA model presented a smaller MAE value, which implies that its projections were very close to the actual sales values than Prophet.

Root Mean Squared Error (RMSE)

Root Mean Squared Error is the square root of average squared errors between the actual and the predicted value. RMSE punishes large errors more severely and it is therefore useful in identifying models that commit severe prediction errors.

Results:

ARIMA RMSE: 7,508.69

Prophet RMSE: 8,136.35

ARIMA model also shows a small value of prediction error and this indicates that it can be used to give more stable and accurate forecasts.

R^2 Score

The R2 can be used to measure the extent to which a model accounts for the variation in the data. An increase in the value is better performance of the model.

Results:

ARIMA R^2: −3.7516

Prophet R^2: −4.5792

Both models produced negative R^2 scores, which warrants critical reflection. A negative R^2 indicates that the models performed worse than a simple horizontal mean baseline on the test data. This outcome is primarily attributable to three factors: first, the limited training data of only 28 months may be insufficient for capturing long-term seasonal patterns; second, the

high month-to-month volatility in the sales data makes precise prediction inherently difficult; and third, the test period (May–December 2023) may contain irregular patterns not well represented in the training period. Despite both models yielding negative R^2 values, ARIMA demonstrated comparatively better performance across all three metrics (MAE, RMSE and R^2), suggesting greater stability and reliability for this specific dataset. These results highlight the importance of acquiring longer historical data and potentially incorporating external variables such as promotions, holidays and economic indicators to improve future model performance

13.3 Model Comparison

The comparison of the two forecasting models shows clear differences in their performance.

Metric	ARIMA	Prophet	Better Model
MAE	6,653.64	7,250.84	ARIMA
RMSE	7,508.69	8,136.35	ARIMA
R^2 Score	-3.7516	-4.5792	ARIMA

These results clearly indicate that the **ARIMA model produces more accurate forecasts than the Prophet model**, as it consistently achieves **lower MAE and RMSE values and a relatively better R^2 score**, indicating smaller prediction errors and better overall forecasting reliability.

In general, the ARIMA model has worked better than Prophet on all performance measures and therefore has made more accurate predictions on this retail sales data.

Among other factors, the high performance of ARIMA is partly due to the fact that the data presented in the dataset has relatively stable trends that are well represented by statistical time-series models. ARIMA gives clean forecasts as per the long-term trends in data.

Prophet on the other hand is engineered to embrace intricate seasonal trends and unexpected variations in trends. Although Prophet was able to recognize the seasonal differences in the sales data, its forecasts were more unstable and it also led to slightly greater forecasting errors.

Nonetheless, Prophet remains an important source of material since the model was able to establish seasonal trends, as well as coming up with confidence intervals that reflect uncertainty in the forecasts.

13.4 Forecast Result Discussion

The forecasting outcomes give significant information on what the future sales performance of the retail business will be.

Because the ARIMA model produced lower forecasting errors, businesses can rely more on ARIMA predictions for operational planning such as inventory control, staffing decisions, and short-term sales forecasting.

The ARIMA forecast shows that the sales per month will merely stabilize at 40,880 units during the forecast period, where the increase will be very minimal with the period.

This indicates that the company can have comparatively steady demand in the near future. Constant demand is good in the operational planning since companies can be able to have the same amount of inventory and be able to allocate resources effectively.

Prophet model, however, shows more seasonal patterns of sales figures. The forecasts indicate that months can have more than low sales and demand, and vice versa. Considering this, the model predicts that the sales are high in some months like June and low in some months like September.

By integrating the insights of the two models, one can have a more accurate picture of how the level of sales is expected to perform in the future:

- ARIMA underlines the stability in sales over the long run.
- Prophet uncovers the seasonal demand variations.

The insights enable the businesses to anticipate future changes in the market and enhance operational decision-making.

13.5 Dashboard and Visual Analytics.

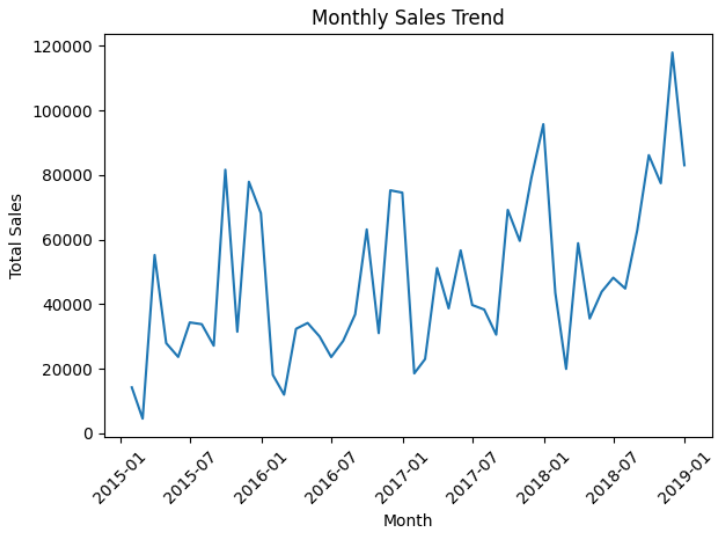
The visual analytic tools were applied to display the results in interactive dashboards to simplify the interpretation of the forecasting results. In this project, **Microsoft power BI** was used to create dashboards.

Visual dashboards enable business users to interactively view the complex analytical outputs in easy to understand visualisations.

The sales forecasting dashboard has various important visual elements.

Historical Sales Trend

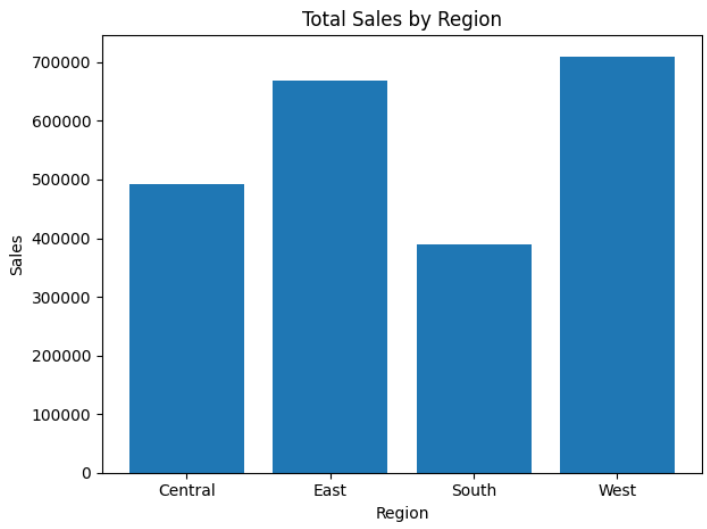
A line chart is used to present the historical sales trend of the company and indicates the way how the values of the sales change with time.



With this visualisation, an analyst can attribute patterns, seasonal changes, and trends in sales data. It is imperative to understand historical trends to create correct forecasting models.

Regional Sales Performance

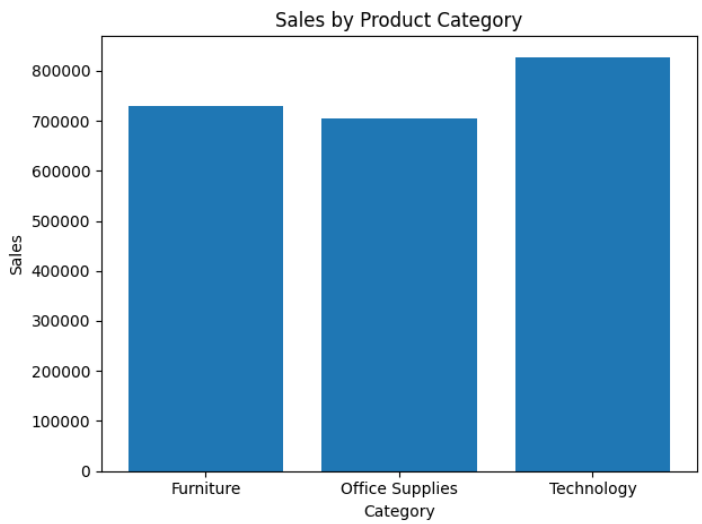
A bar chart is used to compare sales performance across different geographical regions.



The chart brings out the areas that bring in the most revenue. This data assists the business to know good markets and where further marketing plans might be needed.

Product Category Analysis

Another bar chart displays total sales by product category.



This visualisation can be used to determine which product segments are bringing the highest level of overall revenue. This information can help businesses to focus on the successful product category and optimise inventory.

Forecast Visualisation

There are also the dashboard charts of forecasts that indicate the predicted values of sales as generated by the ARIMA and Prophet models. These charts enable the decision makers to compare the past sales performance to forecasts in future.

Interactive options that include filters and slicers allow a user to analyse the information in respect to particular areas, products, or time. This flexibility enables the business managers to delve deeper into the data.

13.6 Business Insights

There are a number of valuable advantages of sales forecasting to the retail and e-commerce organisations.

Inventory Management

Proper predictions enable companies to keep their stocks at the best levels. The ability of companies to forecast demand in the future can make sure that products are accessible when they are required by the customer without having to store large inventories at high costs.

Supply Chain Planning

Sales forecasting also enhances coordination of the supply chain. Businesses are able to plan production, transport, and distribution operations according to the patterns of expected demand, which decreases the inefficiency in the operations.

Marketing Decision-Making

The insights of forecasting enable the marketing departments to plan the promotional activities to take place at the time of peak demand. As one example, when it is forecasted that you will sell more during particular months of the year, marketing campaigns can be scheduled to coincide with those months in order to get the highest sales out of it.

14. Conclusion

This project successfully demonstrated the application of two time-series forecasting models, ARIMA and Prophet, to predict retail sales performance. The comparative analysis revealed that ARIMA outperformed Prophet across all three-evaluation metrics (MAE: 6,653.64 vs

7,250.84; RMSE: 7,508.69 vs 8,136.35; R^2: −3.7516 vs −4.5792), making it the more reliable model for short-term operational forecasting in this context.

However, it is important to critically acknowledge the limitations of this study. Both models produced negative R^2 scores, indicating that neither model achieved strong predictive accuracy against the test data. This suggests that 28 months of training data may be insufficient for robust time-series forecasting, and that the high volatility inherent in retail sales data presents a significant modelling challenge. Additionally, neither model incorporated external variables such as promotional events, public holidays, or macroeconomic indicators, which are known to significantly influence retail sales patterns.

Despite these limitations, the project provides valuable directional insights for business decision-making. ARIMA forecasts are recommended for short-term operational planning such as inventory management and staffing, while Prophet's seasonal decomposition offers useful strategic insights for marketing campaign planning and identifying peak demand periods.

Future work should consider several improvements: first, extending the historical dataset to at least 3–5 years of real transactional data; second, incorporating external regressors into the Prophet model to capture promotional and seasonal effects; third, exploring ensemble approaches that combine ARIMA and Prophet predictions to leverage the strengths of both models; and fourth, investigating machine learning approaches such as LSTM (Long Short-Term Memory) neural networks, which have demonstrated strong performance in complex retail forecasting scenarios.

Overall, this project demonstrates that data analytics solutions combining statistical forecasting models with interactive visual dashboards can effectively transform raw sales data into actionable business intelligence, supporting more informed and strategic organisational decision-making.

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